



Machine Learning, Artificial Neural Networks and Deep Learning

Module 1 (Machine Learning)

Alessandro D'Amelio
Matteo Papini



WHAT (INTRODUCTION)

Full Course: Machine Learning, Artificial Neural Networks and Deep Learning
@ Bachelor of Science in Artificial Intelligence

Module 1: Machine Learning

Instructors:

- Dr. Alessandro D'Amelio
- Dr. Matteo Papini

1st Semester

UNIPV

Module 2: Artificial Neural Networks and Deep Learning

Instructors:

- Prof. Marco Frasca

2nd Semester

UNIMI(b)

WHEN AND WHERE

MONDAY

from 16:00 to 17:30

No breaks

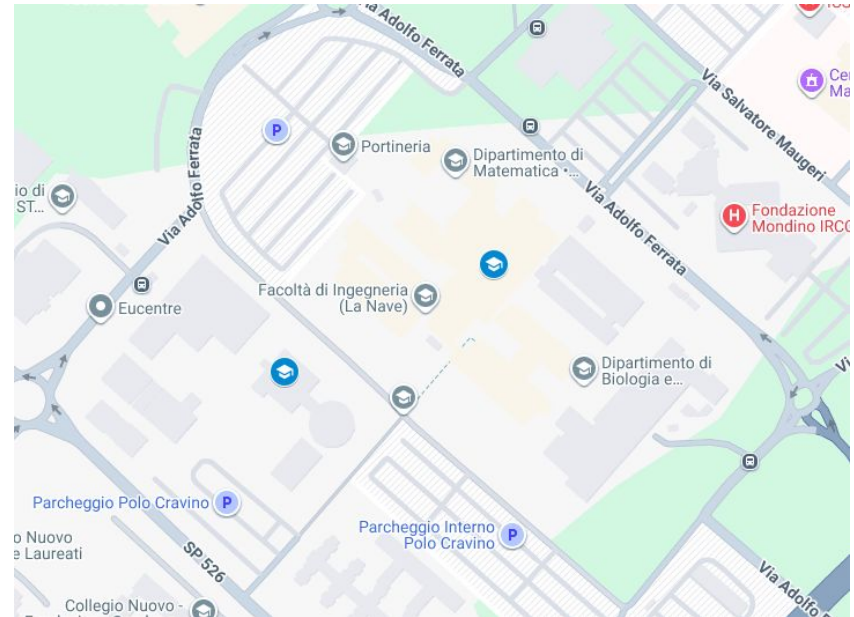
Lab C2/C3

TUESDAY

from 09:30 to 12:00

15' break

Room 4



WHO (Module 1)

Dr. Alessandro D'Amelio

email: alessandro.damelio@unimi.it

Office hour:

- appointment by email
- online on Zoom/Meet



Dr. Matteo Papini

email: matteo.papini@unimi.it

Office hour:

- appointment by email
- online on Zoom/Meet





Course Topics

1. **Introduction** to machine learning and learning tasks;
2. **Classification Models:** Perceptron, Logistic Regression, (kernel) Support Vector Machines, Decision Tree;
3. **Ensemble learning:** bagging, Random Forest, XGBoost, boosting;
4. **Data preprocessing:** Dealing with missing data, handling categorical data, feature scaling, feature selection;
5. **Dimensionality reduction:** PCA, LDA, kernel PCA;
6. **Model evaluation and hyper-parameter tuning;**
7. **Clustering algorithms:** k-means, hierarchical clustering and density-based clustering.

GOALS

Understand the **theoretical concepts** behind the main and most spread **machine learning algorithms** and **apply** them in **different** application **contexts**



Classify a **learning task** according to the input and the output types



Design and implement a **machine learning-driven project** from the initial steps: data preparation and data pre-processing to model evaluation and testing



Think critically about the **usage** of machine learning algorithms in solving a specific learning task



PREREQUISITES

Knowledge of **basic computer science principles** and **skills**
Write a non-trivial computer program in Python



Familiarity with **probability theory**

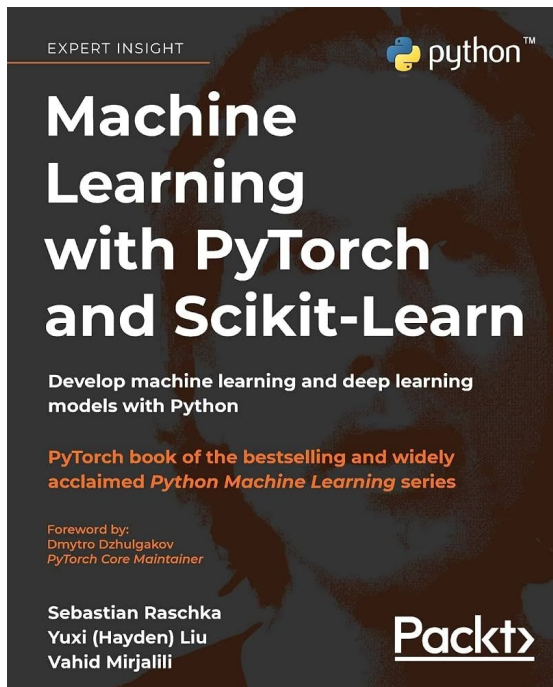


Familiarity with **linear algebra**



COURSE MATERIALS

- Sebastian Raschka & Vahid Mirjalili, “Python Machine Learning”, Packt Publishing, 3rd Edition, 2019
- Sebastian Raschka, Yuxi Liu & Vahid Mirjalili, “Machine Learning with PyTorch and Scikit-Learn”, Packt Publishing, First Edition, Febr. 2022
- Aurelien Geron, “Hands-on Machine Learning With Scikit-learn, Keras, and Tensorflow: Concepts, Tools, and Techniques to Build Intelligent Systems”, O’Reilly, 3rd Edition, Oct. 2022
- Andriy Burkov, “The Hundred-Page Machine Learning Book”, First Editin, Jan 2019
- Andreas Mueller & Sarah Guido, “Introduction to Machine Learning with Python”, O’Reilly; 1st Edition, Oct. 2016



Exam and Grading

Module 1 (ML)

- A **home project** on a learning task **proposed** or **selected** by the student.
- It will be **individually** developed in **Python**.
- **Presentation** of the project and an **oral discussion** starting from the project.
- **Criteria:** a) quality of the project, b) design choices, and c) mastery of the concepts

X/30

Module 2 (NN & DL)

- A **home project** on a **restricted** set of **topics selected by the teacher**.
- It will be developed in **Python**.
- Individually or in teams of up to two members
- **Presentation** of the project and an **oral discussion** of the project.
- **Criteria:** a) quality of the project, and b) mastery of the concepts

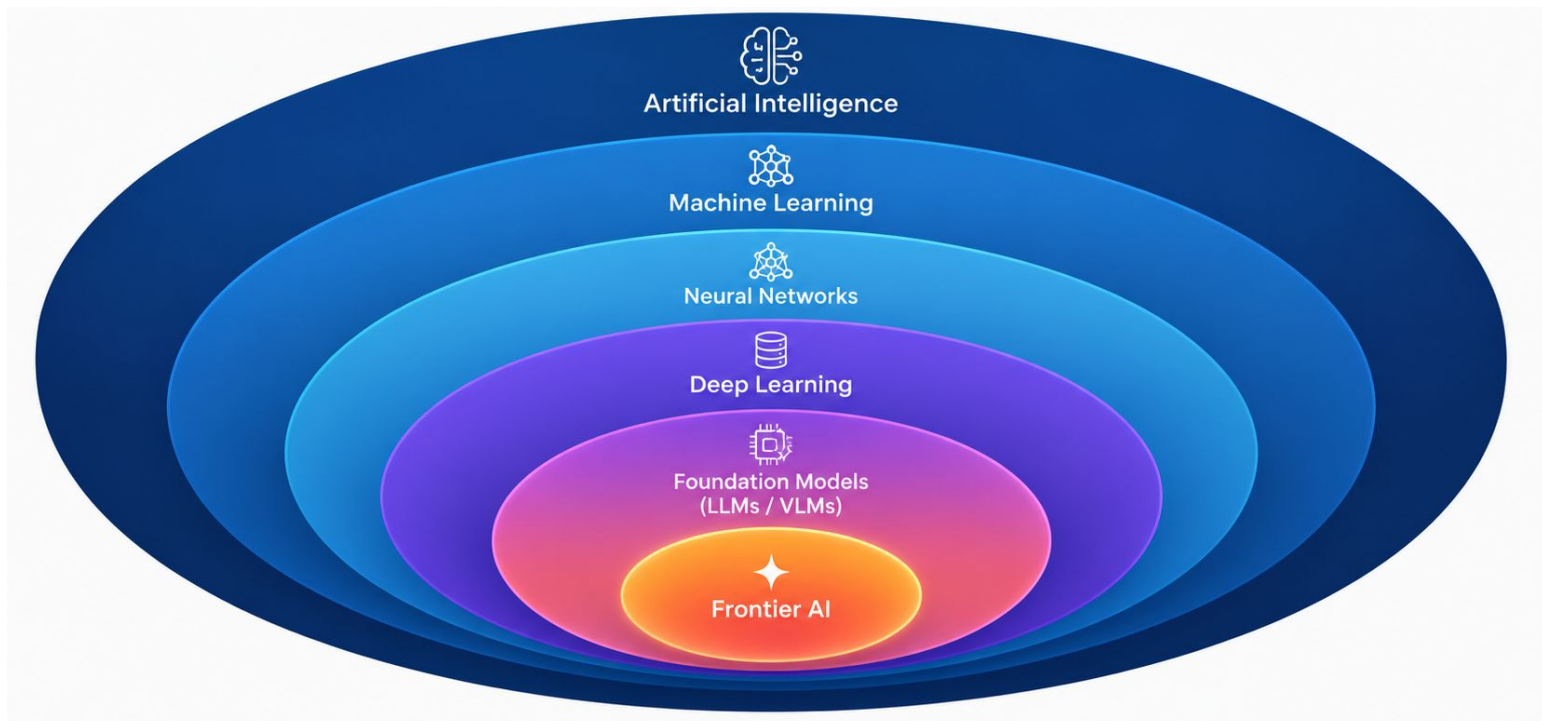
Y/30

+

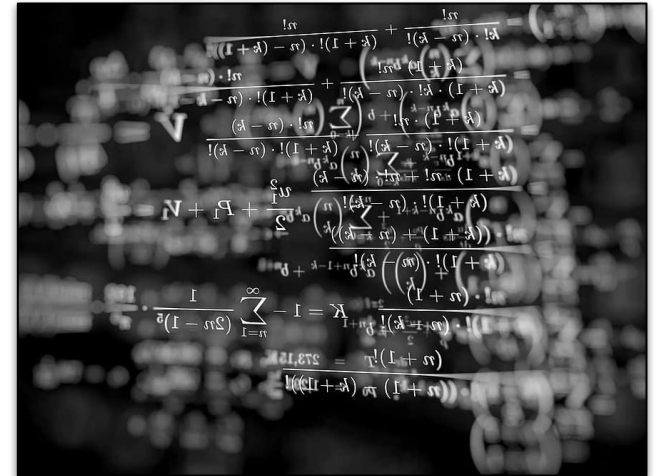
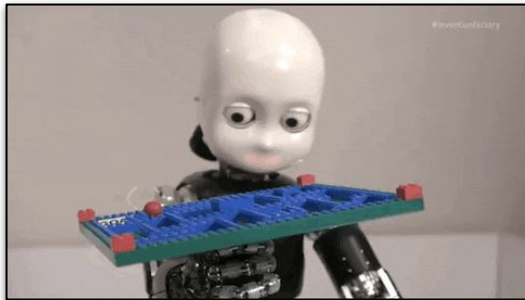
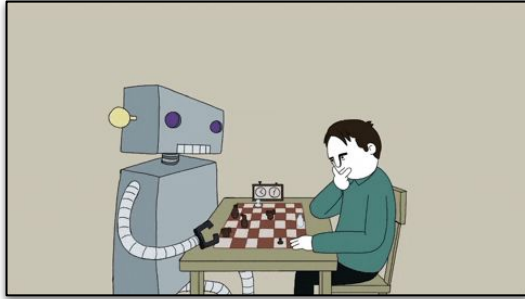
Final Grade Z/30

Machine Learning

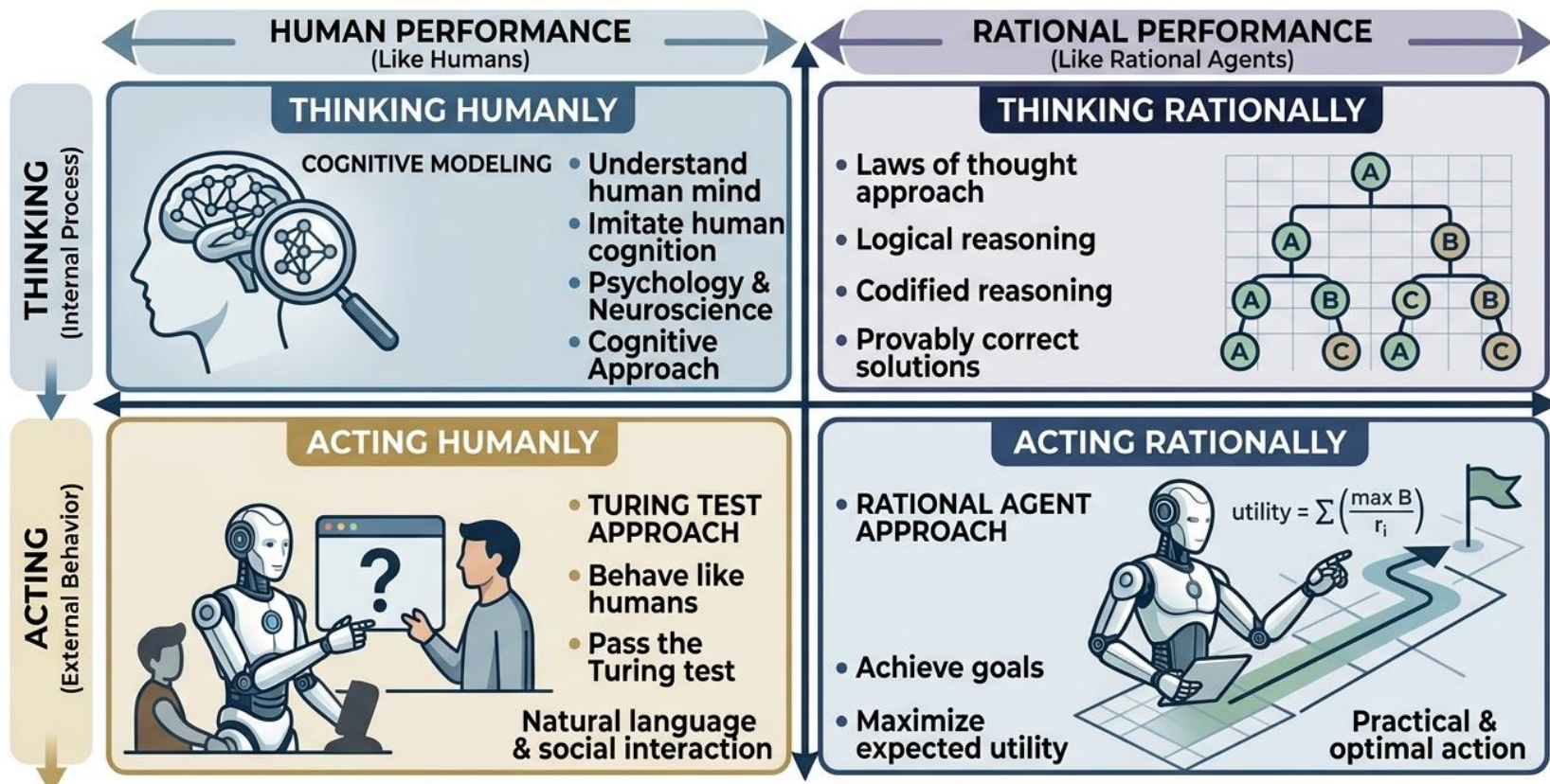
Machine Learning... It has to do with AI, right?



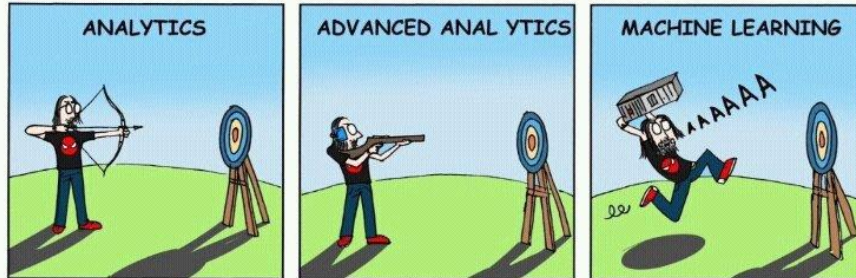
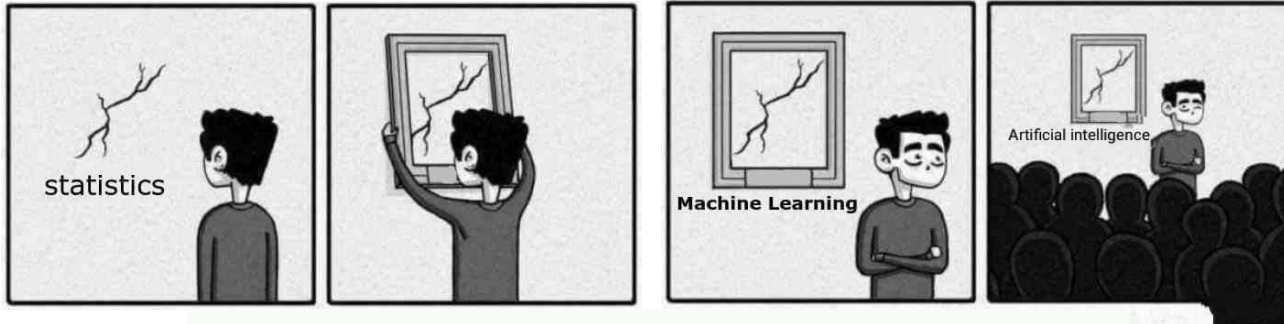
What is (Artificial) Intelligence?



What is Artificial Intelligence (S. Russell and P. Norvig)



What is Machine Learning?



SM. MAR 2017.



What is Machine Learning?



Arthur Samuel, 1959

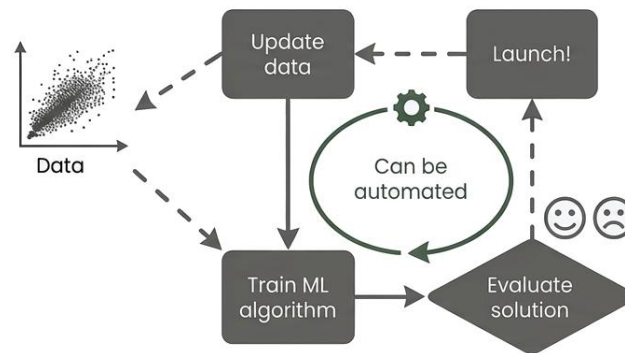
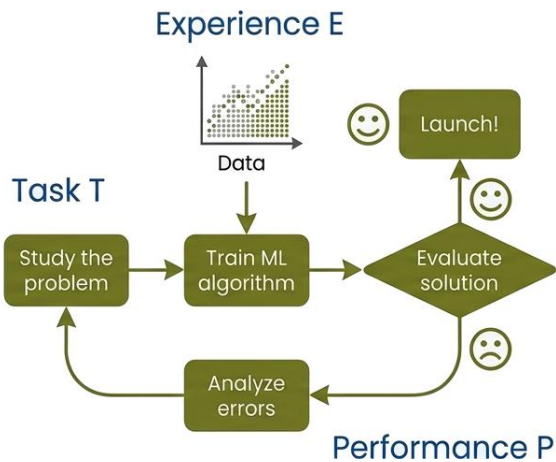
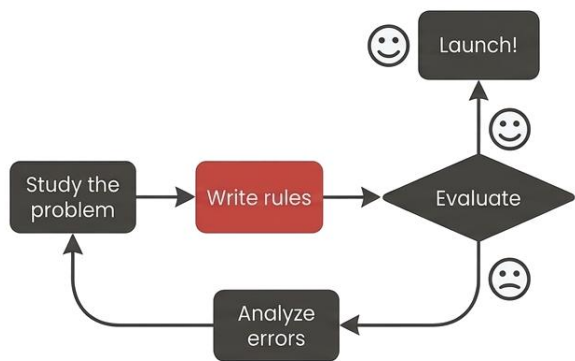
Machine Learning is the field of study that gives **computers** the ability to **learn without** being **explicitly programmed**

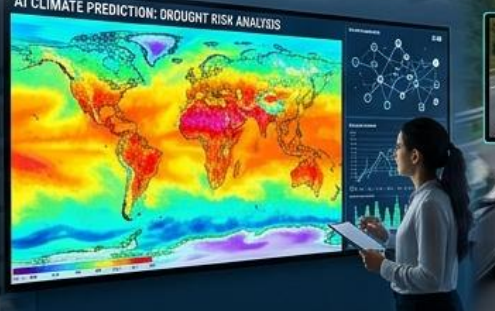
A computer program is **said to learn** from **experience E** w.r.t. some **task T** and some **performance** measure **P**, if its performance on T, as measured by P, **improves with** experience **E**



Tom Mitchell, 1997

Why Machine Learning?





History of Machine Learning

1. Foundations

1943 – 1955

1943 – 1956

Neural & machine-intelligence foundations

- McCulloch–Pitts neuron
- Turing's imitation game
- Dartmouth workshop

2. Symbolic AI

1956 – 1973

1957 – 1969

Perceptrons, symbolic programs & early limits

- Perceptron
- ELIZA
- XOR limitation

3. Expert Systems & AI Winter

1970s – 1980s

1972 – 1980

Expert systems & first AI winter

- MYCIN
- First AI winter

4. Connectionism & Statistical ML

1986 – 2005

1986 – 1997

Learning algorithms & machine game-playing

- Backpropagation
- Deep Blue

5. Deep Learning

2006 – 2016

2006 – 2012

Deep learning renaissance

- Deep Belief Nets
- ImageNet
- AlexNet

6. Transformers & Foundation Models

2017 – 2021

2017 – 2020

Attention, transformers & generative modeling

- Transformer
- Diffusion models

7. Generative, Multimodal, Reasoning & Agents

2022 – 2026

2023 – 2026

Foundation models, multimodality, reasoning & agents

- GPT-4
- Claude 3/3.5
- DeepSeek-R1
- Qwen3
- Frontier AI

"Types" of Machine Learning



Supervised Learning

Learns from data with known answers (labels).

Labeled data

Input (x)	Label (y)
	cat
	dog
	car
...	...



Prediction



Typical goals

Predict the correct answer for new data.



Common tasks

- Regression (predict a number)
- Classification (predict a category)



Example

Predict house prices from features like size, location and number of rooms.



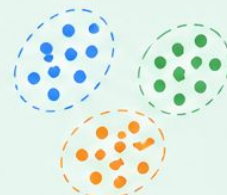
Unsupervised Learning

Learns from data without labels, finding patterns or structure.

Unlabeled data



Discovered structure



Typical goals

Find hidden patterns, groups or representations in the data.



Common tasks

- Clustering (find groups)
- Dimensionality reduction (compress data)
- Anomaly detection (find unusual data)



Example

Group customers by behavior (e.g., spending and purchase history).

Beyond (Un)Supervised Learning



Semi-supervised learning

Uses a small amount of labeled data + lots of unlabeled data.

Example: pseudo-labeling, consistency training



Self-supervised learning

Creates learning signals from the data itself.

Example: masked prediction, contrastive learning



Reinforcement learning

Learns through interaction and rewards.

Example: games, robotics, sequential decision making



Imitation / demonstration learning

Learns from examples of expert behavior.

Example: behavior cloning, imitation learning



Preference / human-feedback learning

Learns from preferences or feedback instead of exact labels.

Example: RLHF, reward modeling, preference optimization



Knowledge distillation & teacher-student

Transfers knowledge from a larger model (or teacher) to a smaller one.

Example: DistilBERT, DINO, BYOL



Generative modeling

Learns to generate new data or model the data distribution.

Example: GANs, diffusion models, VAEs



Transfer learning

Reuses knowledge from one task or domain to another.

Example: ImageNet → medical imaging, pretraining + fine-tuning



Meta-learning

Learns how to learn (fast adaptation to new tasks).

Example: MAML, few-shot learning



Parameter-efficient adaptation

Adapts large models by updating only a small number of parameters.

Example: LoRA, adapters, prompt tuning



Specialized training settings

Different ways to organize or distribute the training process.

Example: active learning, continual learning, federated learning

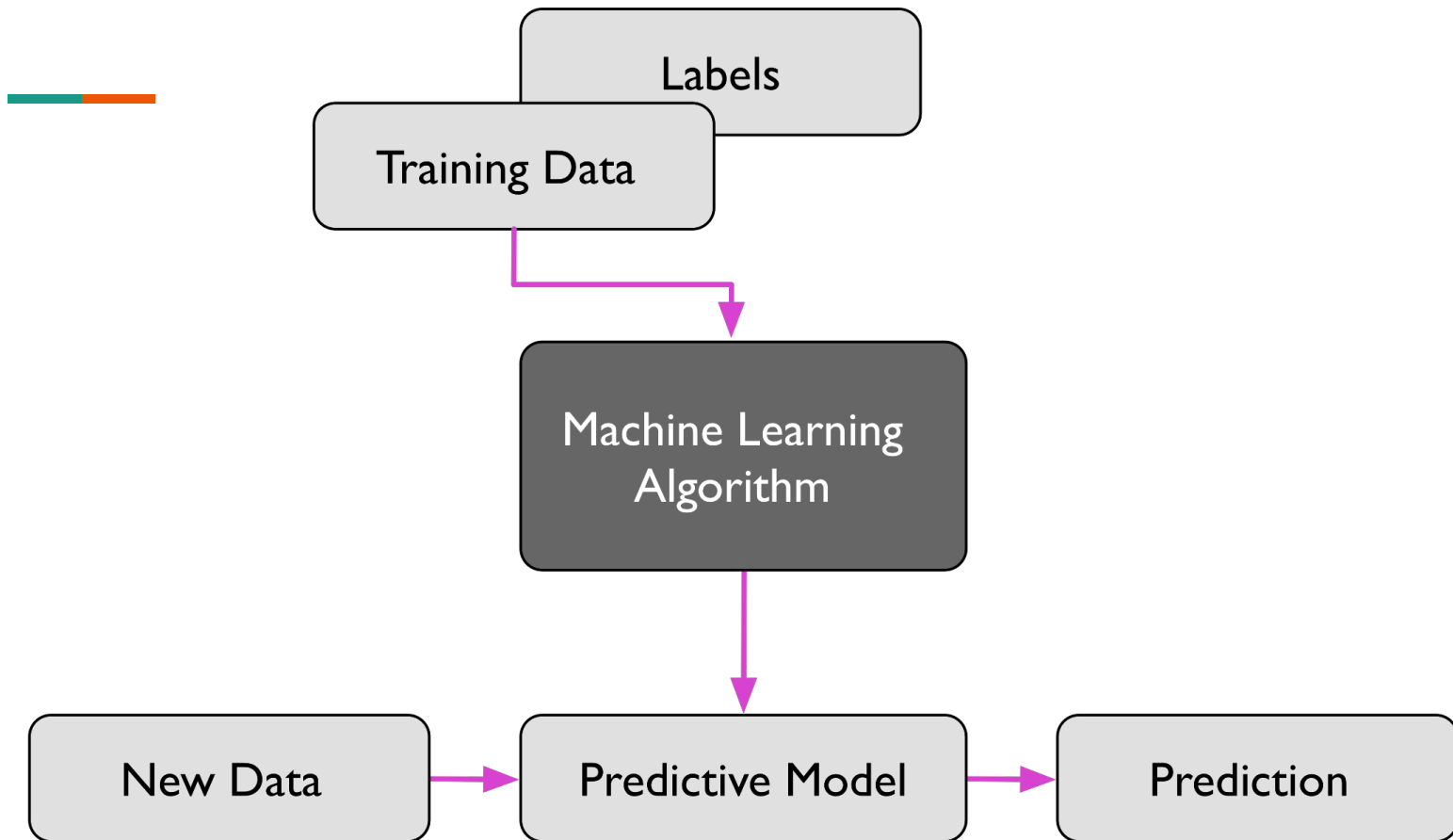


Structured / relational learning

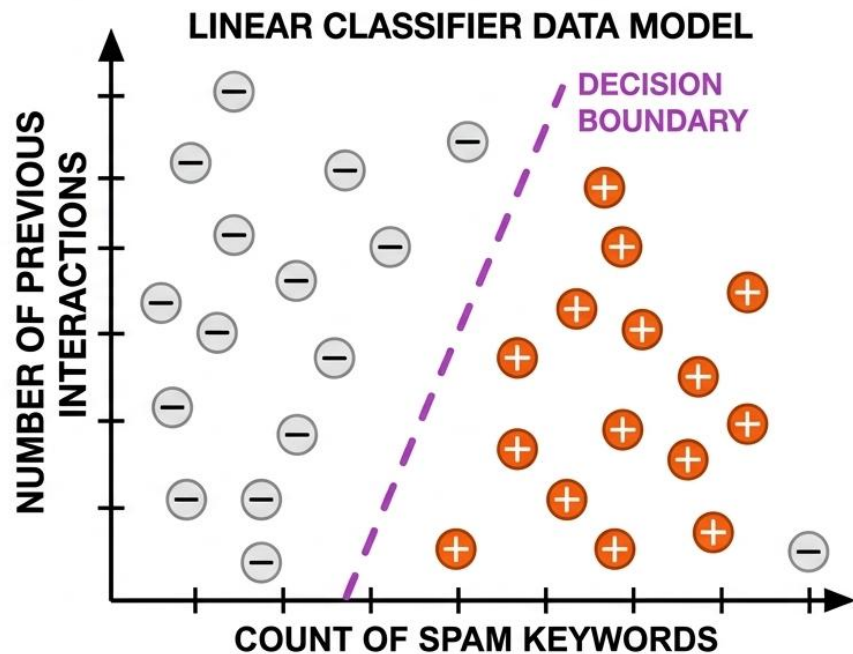
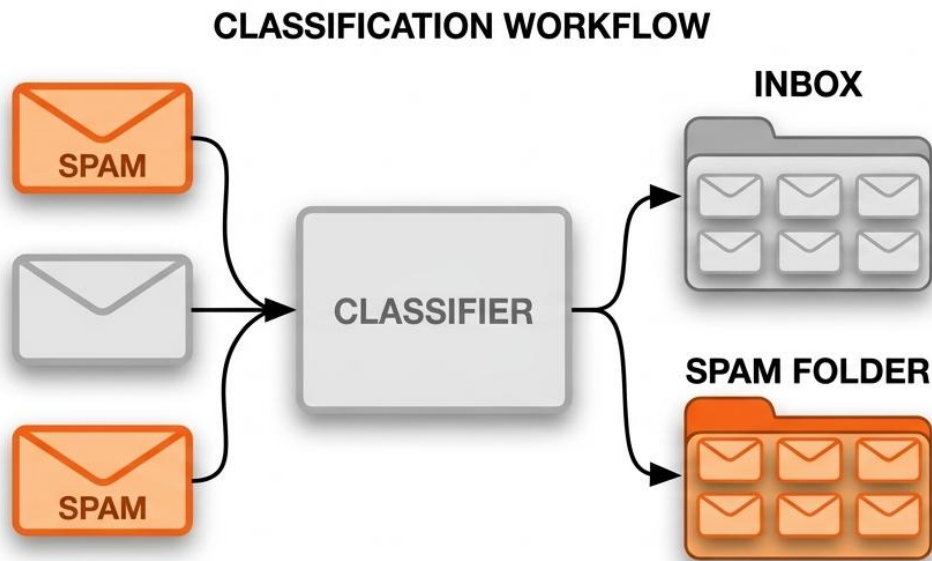
Models data with explicit structure (e.g., graphs, sequences, space).

Example: GNNs, sequence models, structured prediction

Supervised Learning

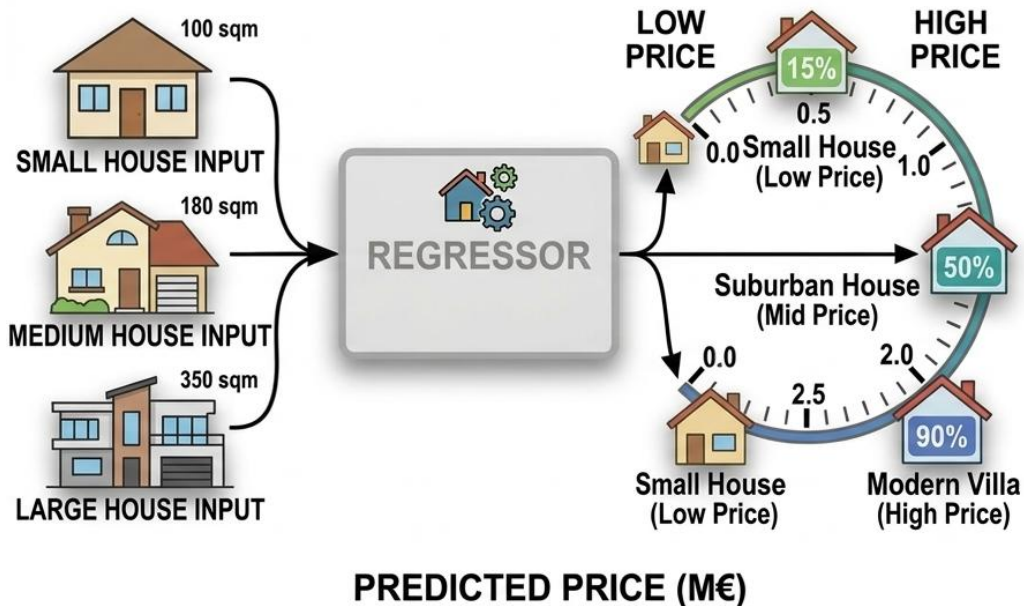


Supervised Learning - Classification

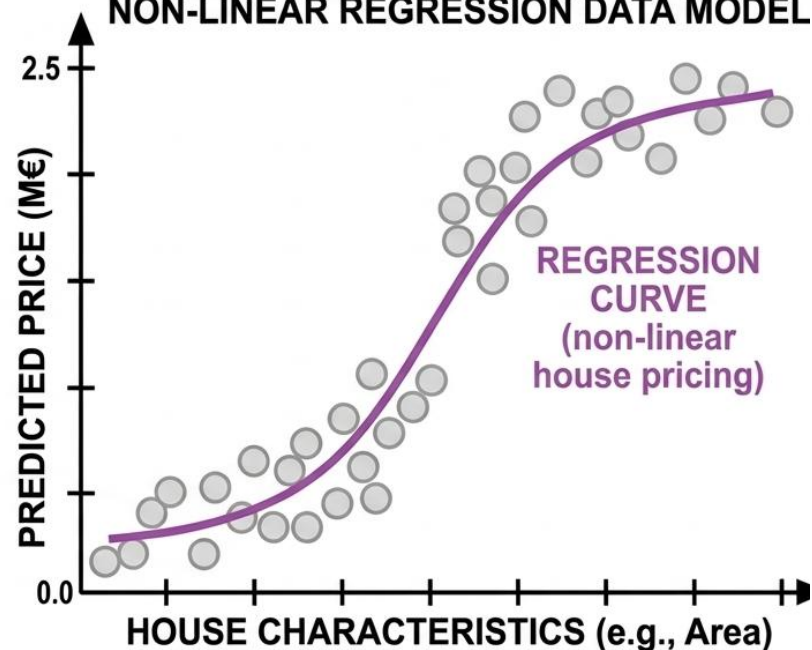


Supervised Learning - Regression

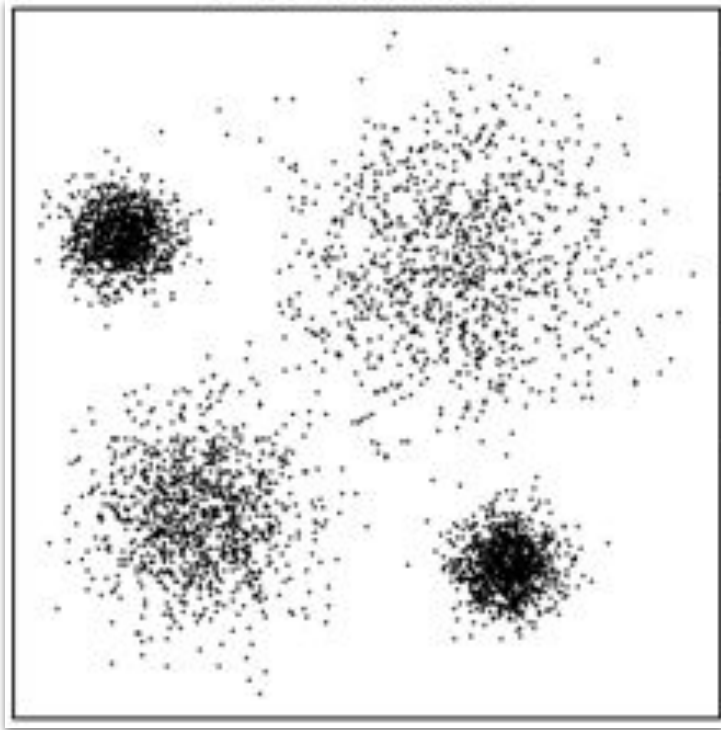
REGRESSION WORKFLOW



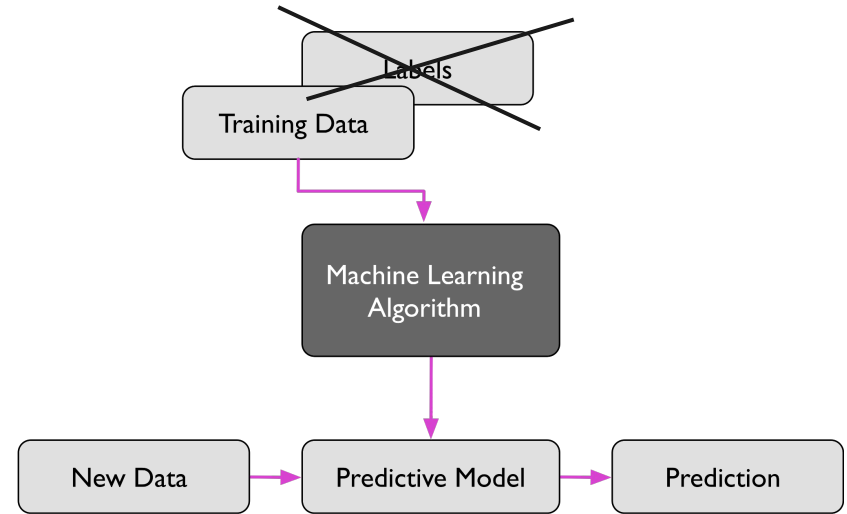
NON-LINEAR REGRESSION DATA MODEL



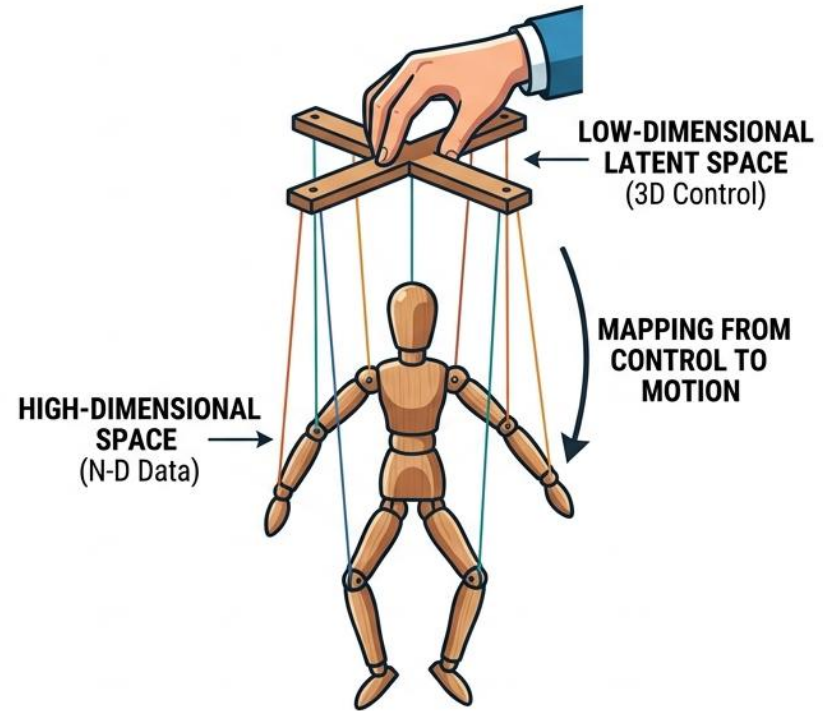
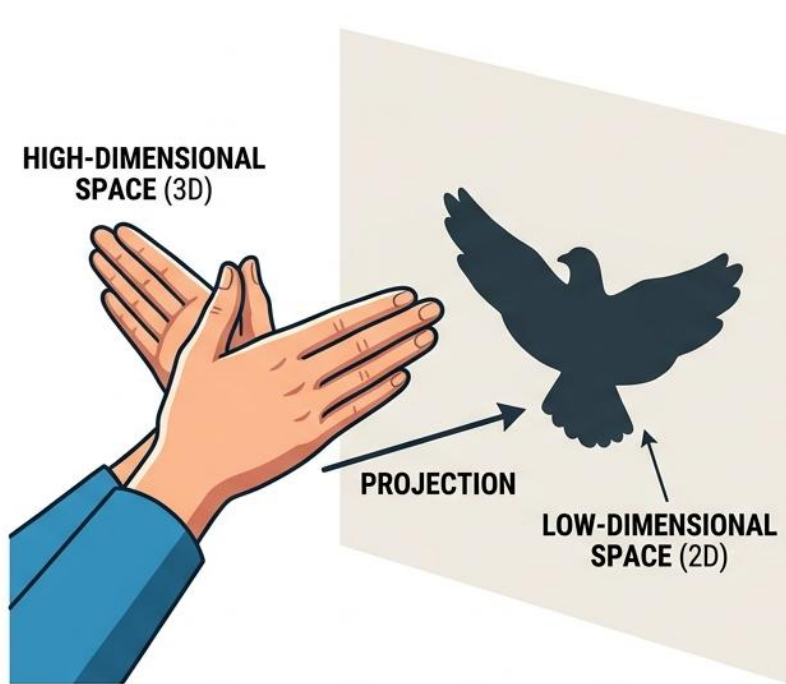
Unsupervised Learning - Clustering



Finding “structure” in the data

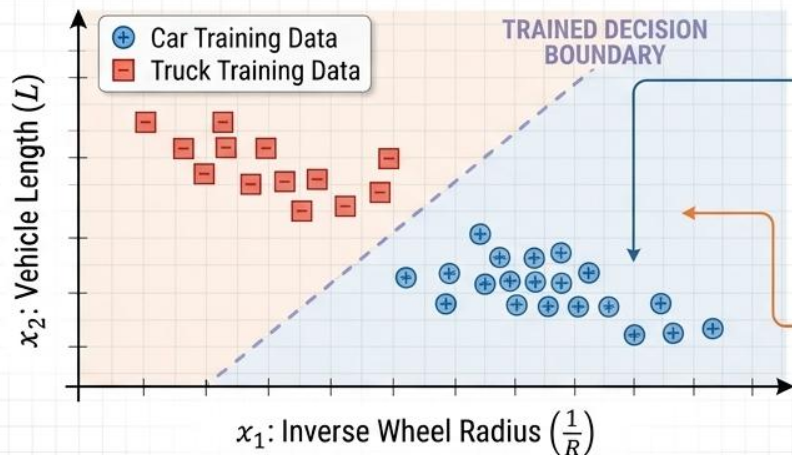


Unsupervised Learning - Dimensionality Reduction



The “Machine Learning Pipeline” (Classification)

TRAINING PHASE VISUALIZATION (OFFLINE LEARNING)



TRAINING EXAMPLES: CAR CLASS



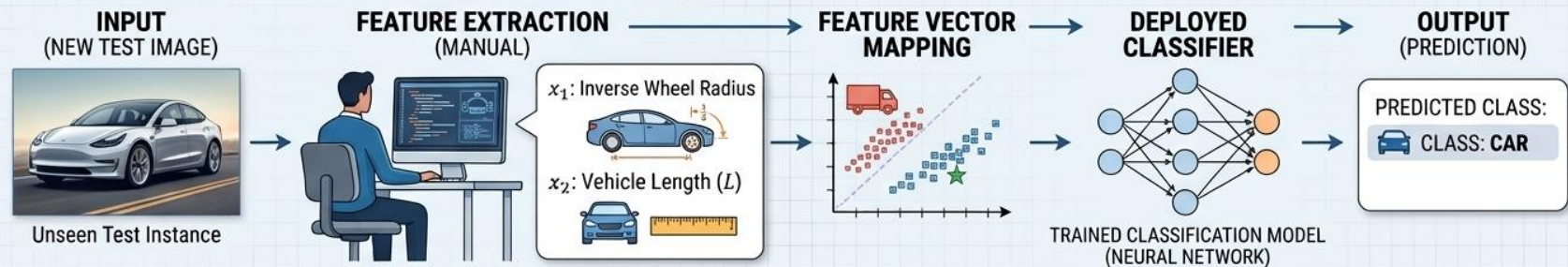
CAR FEATURES: Smaller length, Smaller wheels (larger $\frac{1}{R}$)

TRAINING EXAMPLES: TRUCK CLASS



TRUCK FEATURES: Larger length, Larger wheels (smaller $\frac{1}{R}$)

INFERENCE PHASE PROCESS FLOW (ONLINE PREDICTION)



The “Machine Learning Pipeline” - The IRIS dataset

Samples

(instances, observations)

	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica

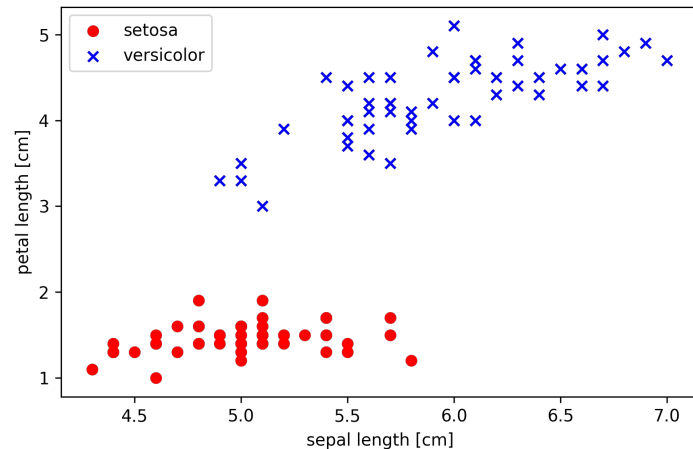
Features

(attributes, measurements, dimensions)

Petal

Sepal

Class labels
(targets)



Machine Learning - The big picture

